

Abin Anil

Python Backend Engineer | REST APIs, Async Services & Cloud Deployment

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Professional Summary

Python backend developer who has independently built asynchronous REST APIs (FastAPI, Django), resolved asyncio concurrency failures, and deployed containerized services to GCP Cloud Run. Backed by 2+ years of enterprise Technical Support Engineering – API debugging, SQL, observability (Kibana, Grafana), incident response, and root-cause analysis on a live production platform sustaining 99.8% uptime. Experienced with PostgreSQL/pgvector data layers, Docker, and Jenkins CI/CD.

Technical Skills

Programming: Python, SQL, JavaScript, OOP, Data Structures & Algorithms

Backend Development: FastAPI, Django, REST API Design, Asynchronous Programming (asyncio), Concurrency & Multi-Threading

Databases: PostgreSQL, pgvector, SQLAlchemy, pycopg, Hybrid Search (BM25 + Vector Indexing/HNSW)

Cloud/DevOps: Docker, Docker Compose, GCP Cloud Run, Jenkins CI/CD, Git, GitHub

Reliability/Observability: API Debugging, Root Cause Analysis, Incident Management, Kibana, Grafana, Postman, Swagger, Network Traffic Inspection

Frontend Integration (working knowledge): React, Vite, Tailwind CSS, Recharts

Applied ML/AI (domain exposure): Python-based ML pipelines, Hugging Face, LangChain/LangGraph service integration, Vector Embeddings, Retrieval-Augmented Generation (RAG), Generative AI, Generative AI Agents, Large Language Modeling, LLM Application Development, Tool Calling, Prompt Engineering, Prompt Patterns, Context Engineering – used as backend building blocks, not the core specialization

Projects

Multi-Agent Service Orchestration Platform (Nexus Intelligence)

Sep 2026

GitHub: <https://github.com/abinanil000datascience/nexus-ai-trading-engine>

- Rebuilt a single-threaded Streamlit prototype into a fully decoupled, asynchronous backend on FastAPI, exposing four independent service workers (Scout, Quant, Analyst, Gatekeeper) behind a unified API layer.
- Diagnosed and eliminated critical asyncio event-loop collisions between FastAPI and background execution threads, restructuring endpoints to isolate long-running workloads without blocking the request/response cycle.
- Integrated a real-time data ingestion layer (streaming market data via Pandas) with downstream API consumers, processing data continuously alongside incoming API requests.
- Built and served a REST API layer for a React/Vite/Tailwind CSS dashboard, delivering live service telemetry and computed results to the frontend in real time.

Tech Stack: Python, FastAPI, asyncio, Pandas, REST APIs, React, Vite, Tailwind CSS, Recharts.

IT Ticket Resolution Backend (Healthcare ITSM Engine)

Aug 2026

GitHub: https://github.com/abinanil000datascience/Healthcare_ITSM_Agent

- Designed a sub-linear hybrid search architecture in PostgreSQL/pgvector, combining keyword matching (tsvector/BM25) with HNSW-indexed vector search to accelerate retrieval across a historical ticket corpus.
- Built and deployed an asynchronous FastAPI backend serving retrieval and classification results, fully containerized with Docker for reproducible deployment.
- Eliminated ongoing third-party API costs by replacing a cloud-dependent inference call with a self-hosted local model service, cutting external dependencies from the request path.

Tech Stack: PostgreSQL, pgvector, HNSW, FastAPI, Docker, Python, Hugging Face.

Applicant Matching Platform Backend (Enterprise AI Career Platform)

Aug 2026

GitHub: <https://github.com/abinanil000datascience/ai-career-platform>

- Built an asynchronous FastAPI backend with a PostgreSQL/pgvector data layer to serve resume-matching and applicant-tracking workflows.
- Designed the API and data schema for storing, querying, and ranking structured applicant data.
- Deployed the containerized service to GCP Cloud Run, with Docker-based build and release packaging.

Tech Stack: Python, FastAPI, PostgreSQL, pgvector, Docker, GCP Cloud Run.

Supply Chain Risk API (AI-Powered Supply Chain Risk Predictor)

May 2026

GitHub: <https://github.com/abinanil000datascience/supply-chain-risk-predictor>

- Built a Django REST API backend serving real-time delay-risk predictions to a React front end, on a 97% on-time, class-imbalanced dataset.
- Designed the API contract and response schema to give end users real-time visibility into at-risk shipments.

Tech Stack: Python, Django, REST APIs, React, Scikit-Learn.

Professional Experience

Technical Support Engineer, Feathersoft

Distributed DNA diagnostics platform (DDC)

Aug 2024 – Present

- **API & Database Operations:** Managed production reliability for a distributed DNA diagnostics platform (DDC), ensuring data integrity across complex probability (PI) calculations and automated report generation workflows, sustaining 99.8% system uptime.
- Reduced production error rates by 22% by debugging API-layer failures and data-ingestion bottlenecks through network traffic inspection, cURL requests, and payload validation using Postman and Swagger.
- Spearheaded system observability and anomaly detection by analyzing application logs in Kibana and Grafana to isolate root causes of complex workflow disruptions, accelerating incident resolution time by 15%.
- Reduced Mean Time to Resolution (MTTR) by 20% by executing direct SQL (SELECT/UPDATE) database interventions to resolve live data discrepancies and maintain the structural integrity of diagnostic reporting.
- Engineered and validated temporary diagnostic workflows within staging and test environments, acting as a technical bridge between end-users and the core development team to unblock production.
- Led cross-functional incident triage with US-based enterprise clients and engineering teams, translating complex diagnostic logs into actionable technical fixes, sustaining a 99% CSAT rating across 120+ escalated incidents.

Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering

University College of Engineering, Thodupuzha

Dec 2020 – May 2024

CGPA: 7.33/10.0

Certifications & Training

- **Fundamentals of AI Agents Using RAG and LangChain** – IBM Sep 2026
Credential ID: S0HTSWVNLSTW | [Verify Credential](#)
- **Data Science With AI Professional Certification** – MetaiLearn Nov 2025 – Jun 2026
- **Python Programming & Machine Learning Intensive Program** – Reverttech IT Solutions (1 month) Dec 2024
- **Advanced Machine Learning Workshop** – Techmaghi Training Institute Sep 2022

Achievements

- **Employee of the Month (January 2026)** – Recognized for technical excellence, critical root-cause analysis, and outstanding client communication.